

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: Theory of Computation

Code No: **CSC262** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **and space complexity of Turing Machine.**
11. How conversion of PDA to CFG is done? Illustrate with an example
12. State Arden's theorem. Convert following DFA into its regular expression using Arden theorem.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Describe the extended transition function of NFA. Construct a NFA, using transition table or transition diagram, over $\Sigma = \{0, 1\}$ that accepts the strings having substring 01 and ends with 1. Show the acceptance of 01111.
2. Define CFG. Construct a CFG that generates the language of all palindromes over $\Sigma = \{a, b\}$ that do not contain the substring ab. Show the leftmost derivation and construct the equivalent parse tree for string babbbab.
3. How Turing Machine is used as a computing function? Construct a TM for simulating a function $f(x) = 2x$ for $x = \{1\}^+$. Iterate the TM to compute 111 and generate the output.

Section B

Attempt any Eight questions [8×5=40]

4. Differentiate Kleen closure from Positive closure. Compute positive and Kleen closure over $\Sigma = \{a, b\}$.
5. Design a Mealy machine over $\Sigma = \{1, 0\}$ that generates output 'A' if the input string ends with 11 else output 'B' if the string ends with 00.
6. Construct regular expression over $\{1 \text{ to } 9\}$ that represents
a. strings of even number with length 4 starting with 1 and ending with 8
b. strings starting with odd number and ending with even numbers
7. Prove that the language $L = \{a^n b^n c^n \mid n > 0\}$ is not context free.
8. Construct a PDA that accepts strings over $\Sigma = \{a, b\}$ that contains equal number of a's followed by equal number of b's. Show acceptance of aabb and aaabbb.
9. Describe how multi-stack TM is different from the semi-infinite tape TM?
10. What is intractability? Define time and space complexity of Turing Machine.
11. How conversion of PDA to CFG is done? Illustrate with an example

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12. State Arden's theorem. Convert following DFA into its regular expression using Arden theorem.
- 01 $\rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_1 \rightarrow q_3 \rightarrow q_2 \rightarrow q_2 \rightarrow q_1 \rightarrow q_2$